LEF MODEL · CHAPTER 11

Chapter 11: Craters Without Asteroids — the Physics of Ballistic Flight, Shock Pressure and the Leading/Trailing Asymmetry

Based on Paper 06 — The Craters (Parts A–K: the physics of launch, shock, passage through vapour and morphology at the four longitudes)

01 · THE MYSTERY

The mystery: can material fired from the ground and falling back onto it dig giant craters — and why is it denser behind?

The accepted literature takes it for granted that almost every crater on the Moon's face was dug by an asteroid or comet arriving from outer space at a tremendous speed of 15–20 km/s. But this assumption faces two sharp observational problems that have never been fully explained:

1. **The opposite of what is expected — an excess of craters on the trailing hemisphere (90°E), not the leading one (90°W):** the Moon always turns the same side toward the Earth, so its western half (around 90°W, **the leading hemisphere**) always faces forward in the direction of its orbital motion — just like the windscreen of a moving car — while its eastern half (around 90°E, **the trailing hemisphere**) faces backward. If external asteroids bombarded the Moon, the windscreen (90°W) must collect tens of per cent more craters than the rear window. In fact the exact opposite happens: **the**

trailing hemisphere (90°E) is much denser in large craters (249–252 per million km² against 139–175 on the leading hemisphere)!

2. **The basic physical question:** in the LEF model, about 99% of the craters were created by **returning material** fired out of the feeding openings of the parent body (the early Earth, with a radius of $r_0 = 6,721$ km and a 5.5-hour day) that landed back on the outer shell, which later rose and became the Moon's face. But can material thrown upward at 2–8 km/s come back hard enough at all to produce shock pressures of tens of gigapascals and stamp complex craters?

02 · AN IMAGE

An image: the ball thrown upward, the figure skater and the runner in the rain

Three simple images from everyday life clarify the physics of ballistic flight and the east–west asymmetry:

- **A ball thrown straight up (the landing speed equals the throwing speed):** when a ball is thrown straight up with no air friction, it converts speed into height on the way up, and converts all the height back into speed on the way down — and returns to the hand at exactly the speed at which it was thrown (5 km/s out → 5 km/s back!).
- **A figure skater spreading her arms (the westward deflection):** while the lump rises to a great height, its angular momentum is conserved — so its angular rate of rotation around the Earth's centre slows (exactly like a skater spreading her arms to the sides). The ground beneath keeps turning eastward at a constant rate, so when the lump comes back down to the same radius, the ground has already moved on eastward and it lands **west of the launch point** (at 5 km/s: to a height of 1,932 km, within 28 minutes, and 1,032 km to the west!).
- **A person running into the rain versus a person running away from it (the leading versus the trailing hemisphere):** after the lunar body had risen, its own feeding openings also fired material (at a relatively low speed, up to its escape speed of ~ 2.0–2.3 km/s). On the trailing hemisphere (90°E), the

ground moves away from the falling material and so "catches" it gently from behind (and also gathers the material drifting west from the far side, 180°); while on the leading hemisphere (90°W), the ground races forward into the falling material and meets it **head-on** — like a runner meeting raindrops head-on!

03 · SCIENCE AND MODEL

Science and model: a shock pressure of 45.6 GPa, passage through vapour and the escape limit (10.75 km/s)

Parts J and K of Paper 06 show by exact analytic calculation that internal launching at speeds of 2–8 km/s is fully sufficient to create true shock craters:

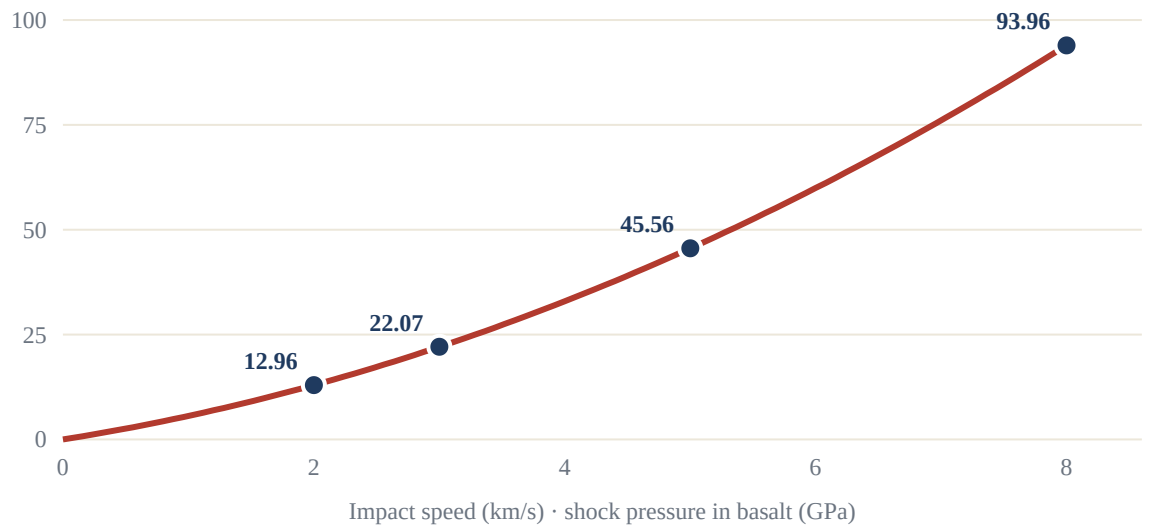
1. **Shock pressures of tens of gigapascals (the Hugoniot equation for basalt, §J.2):** in a head-on collision of basalt with basalt ($\rho_0 = 2,700$ kg/m³), the particle velocity behind the shock front is half the impact speed ($u_p = v_n/2$), and the speed of the shock front itself is $U_s = C_0 + s u_p$ (with $C_0 = 3.5$ km/s and s = 1.3). This gives a contact pressure of **12.96 GPa already at 2 km/s**, **22.07 GPa at 3 km/s**, and **45.56 GPa at 5 km/s** — far above the thresholds of shock metamorphism of minerals! (And even a liquid drop creates a strong compression wave, because a liquid resists compression almost as much as a solid.)

2. **The energy needed at the opening and the escape limit (§J.1, §J.5):** to launch one kilogram at 2, 3, 5 or 8 km/s, a kinetic energy $q_{\text{kin}} = u^2/2$ of 2, 4.5, 12.5 or 32 MJ/kg is needed, supplied by the Earth's internal pressure (passing through the pear) aided by the expansion of volatile gases (water). On the other hand, for the material not to escape to space forever but to come back to the shell, the radial launch speed must be lower than the escape speed combined with the rotation: $u < u_{\text{esc}} \simeq 10.748$ **km/s**.

3. **Crossing the vapour atmosphere (§J.3):** a lump one kilometre across crossing twice (rising and falling) a column of vapour with a base pressure of 100 bar ($g_{\text{eff}} = 8.25$ m/s²) keeps about **51% of its speed**

($v_{\text{out}}/v_{\text{in}} = \exp(-\Sigma_a/2B)$) — so a launch at 5 km/s still lands at 2.5 km/s and produces a shock of more than 13 GPa!

4 **The scaling law under the parent's gravity (§C.6):** since the parent's surface exerted a gravity five times stronger than the Moon's today (g = 8.25 against 1.62 m/s²) and the impact speed was three times lower (5 against 15 km/s), creating a crater of the same diameter requires a returning body **7.45 to 23.47** times heavier ($m_p \propto g^{3\mu/2} v^{-3\mu}$ for $\mu = 0.35$–0.55).



The shock pressure in a basalt-on-basalt impact according to the chapter's Hugoniot equation (ρ_0 = 2,700, C_0 = 3.5, s = 1.3): already at 2 km/s about 13 GPa.

`Splitting the concept of "depth" into three separate heights (§A.2)`

To decode what the craters at the four longitudes are telling us, Paper 06 splits the word "depth" into three independent geometric quantities (at 72 azimuths around each crater, after masking the central peak):

(1) The height of the rim above the surrounding ground — records *the forces acting at the moment of impact*;

(2) The depth of the floor below the surrounding ground — records *the stiffness of the layers beneath the crater*;

(3) The total depth from the crater's rim to its floor — the sum of the first two.

The mechanism in words: the four longitudes — every part of a crater has its own memory

Parts C.3 and F–I of Paper 06 scan 1,457 craters 20–100 km across in four equal caps on the equator (radius 30°, after removing the maria and the SPA basin) and show how each part of a crater records a different aspect of the same process:

1. **The crater count — the trailing hemisphere (90°E) is 1.44–1.79 times denser than the leading one (Study 1, §C.3):** on the trailing hemisphere **249 craters** 20 km or more across were counted per million km² (252 after removing the 20 large basins), against **139** (175) on the leading hemisphere at 90°W and **135** (148) on the near side at 0° (95% confidence interval for the ratio: 1.56–2.05, and in cell sampling 1.48–2.19). The gap is concentrated in the small and medium craters (< 70 km, 1.2–2.4 times), which were created from material fired from the Moon itself and gathered behind, while the giant craters (> 70 km) were created early in the flattened stage and are more evenly spread.

2. **The crater rim (the forces at the moment of impact, Part F):**
    - 3. **On the far side (180°), the highest rims (18.7 × 10^{-3}D at 30–50 km):** the far side faced outward in the common rotation around the centre of mass shared with the Earth, so the centrifugal force pulled the ejected material outward and left high rims (and also high central peaks, 30 × 10^{-3}D).
    - 4. **On the near side (0°), the lowest rims among the large craters (10.3 × 10^{-3}D):** the centrifugal force pressed the material inward against the ground.
    - 5. **On the leading hemisphere (90°W), the rims are jagged, elongated and low at 20–30 km:** the head-on impact shattered and notched the rim (rim irregularity 1.25 times, jumps between azimuths 1.14 times, and elongated rims e > 1.2 in **17%** against 12% at 90°E) and pressed it down (12.7 against 17.2 × 10^{-3}D).

6. **The crater floor below the surface (the stiffness of the layers, Part G.1):**
    - 7. **On the far side (180°), the floor sits deepest (54 × 10^{-3}D at 20–30 km):** the far side is built of soft layers of sial and returning material piled on one another; the glowing material that landed on them crushed and melted them, and the floor sank deep inward.
    - 8. **On the near side (0°), the shallowest floor (27 × 10^{-3}D, half that of the far side):** the stiff ground did not collapse, and the filling of the maria from below raised the floors even further.
    - 9. **Between the leading (90°W) and trailing (90°E) hemispheres there is no difference in floor height:** the head-on impact shaped the rim, the peak and the roughness — but the depth of the floor is set only by the substrate beneath!

10. **The central peak and the roughness of the floor (the impact energy, Parts H–I):** around the transition to complex craters (20–30 km), a head-on impact crosses the energy threshold for forming a central rebound: at **90°W, 60% of the craters have a central peak** (against 46% at 180° and 49% at 90°E), and the floor roughness is **1.50** times higher (4.5° against 3.0°, correlated at r = 0.40 with the notching of the rim) — while in the giant craters (50–100 km), the combined impacts of the Moon and the Earth on the far side produce a peak in **98%** of the craters!



The density of craters 20 km and more across per million km² in four caps on the equator (Study 1, maria and SPA removed): the trailing hemisphere is denser than the leading one.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The laws of conservation of energy and angular momentum on a Keplerian orbit around the parent (equations 10–14 in Paper 06) fix both the impact speed at the same height ($v_{\text{land}} = u$) and the relative impact speed on a higher rotating target ($r > r_0$):

$$E = \frac{u^2 + \omega^2 r_0^2}{2} - \frac{K}{r_0}$$

$$\ell = \omega r_0^2 \quad \Longrightarrow \quad v_{\text{land}}(r_0) = u$$

$$v_{\text{rel}}^2(r) = 2\left(E + \frac{K}{r}\right) - \frac{\ell^2}{r^2} + \left(\frac{\ell}{r} - \omega r\right)^2$$

$$\frac{dv_{\text{rel}}^2}{dr} = 2\left(\omega^2 r - \frac{K}{r^2}\right)$$

The derivative dv_{rel}^2/dr (§K.3) shows that below the corotation radius ($r_{\text{corot}} \approx 15,883$ km with the data of the early stage) gravity dominates, so the relative impact speed on a higher target falls with height ($8.0 \to 6.83 \to 5.49 \to 4.60$ km/s at radii of 6,721, 8,000, 10,000 and 12,000 km), but this is a geometric relation for a given orbit and does not imply a universal decline with time.

06 · THE SIGNATURE

The signature in reality: crater floors standing high above the basalt around them

Studies 10 and 11 of Paper 06 (§G.2–G.3) examine the 86 large craters (40+ km) on the near-side basalt and reveal a phenomenon that side flooding cannot explain:

- **A third of the melted and smeared craters stand above the basalt surface outside them (Study 10):** had external lava flooded a crater from the side, the

level of its floor could never have risen above the level of the basalt outside it (like the law of communicating vessels). In fact, **24 of 76** melted and smeared craters (32%) stand **more than 50 m above the basalt around them** (for example **Plato**, standing +500 m above the basalt of Imbrium, and a crater 182 km across in Oceanus Procellarum standing +518 m above the plain!), against only 12 of 411 (3%) in the highlands!

- **The physics of the frozen wave and the separation of iron (Study 11):** a large lump that landed on a fluid or soft basalt plain bounced a wave upward at its centre; the floor began to sink back, but where cooling won, the uplift froze in place. Indeed, the raised floors (> +50 m) are rich in light volatiles and therefore **poor in iron and titanium** (13.3 wt% FeO, $1.5\ \mathrm{wt\%\ TiO_2}$) compared with the floors that sank down (14.4 wt% FeO, $2.5\ \mathrm{wt\%\ TiO_2}$; significance p = 0.018 and p = 0.010)!

The lab box: the tables of flight, shock, the longitudes and the raised floors (Paper 06, Parts C–K)

Launch speed u (km/s)	Launch energy (MJ/kg)	Peak height above r_0 (km)	Flight time (minutes)	Westward deflection (km)	Landing speed and shock pressure in basalt
2 km/s	2.0 MJ/kg	251 km	8.46 min	51.7 km	2 km/s → **12.96 GPa**
3 km/s	4.5 MJ/kg	591 km	13.50 min	184.1 km	3 km/s → **22.07 GPa**
5 km/s (the reference example)	**12.5 MJ/kg**	**1,932 km**	**28.01 min**	**1,031.6 km**	**5 km/s → 45.56 GPa**
8 km/s	32.0 MJ/kg	8,684 km	97.28 min	8,438.7 km	8 km/s → **93.96 GPa**

Morphological measure (Studies 1 and 6)	90°W (leading)	180° (far)	90°E (trailing)	0° (near)	Physical meaning

Density of 20+ km craters (per million km²)	**139 (175)**	**203 (221)**	**249 (252)**	**135 (148)**	**The trailing hemisphere gathered the ejecta of the Moon and the far side (ratio 1.79, interval 1.56–2.05)**
Central peak at 20–30 km / 50–100 km	**60%** / 94%	46% / **98%**	49% / 81%	56% / 68%	A head-on impact crosses the rebound threshold at 90°W
Rim irregularity / azimuth jump (20–30 km)	**9.2 / 3.9**	7.5 / 3.5	7.4 / 3.5	7.1 / 3.4	1.25 and 1.14 times at 90°W; elongated rims 17% against 12%
Floor roughness / flat area (20–30 km)	**4.5° / 4%**	2.7° / 14%	3.0° / 11%	3.6° / 9%	The impact shock roughens at 90°W (r=0.40 with rim notching)
Rim height above the ground (30–50 km, × 10^{-3}D)	14.4 (12.7 at 20–30)	**18.7**	13.1 (17.2 at 20–30)	13.6 (10.3 in the large ones)	The far side faced outward in the common rotation: the centrifugal force lifted the ejected material and the rims stand high; on the near side the same force pressed the material against the ground

07 · NEXT

The door ahead

In this chapter we saw that Earth material fired from the ground and returning to it at 2–8 km/s produces shock pressures of tens of GPa — more than enough to dig giant craters without any asteroid. We also saw that every part of a crater keeps its own memory: the count shows that the trailing hemisphere is 1.44–1.79 times denser than the leading one, the rim records the forces at the moment of landing (high on the far side, jagged on the leading hemisphere), and the floor records the ground beneath. Finally we saw crater floors standing high above the basalt around them — something side flooding cannot create.

But if the craters were created from returning Earth material, the chemistry inside them must bear witness to it. In the next chapter — **Chapter 12: The Chemistry of the Crater Interior, the Foreign Mass That Is Not There, and the Earth as the Last Scatterer** (from Paper 06 — Parts L–Q) — we will examine the interior of the

craters, look for the foreign mass that asteroids would have had to leave in the Moon's crust, and see how the Earth kept scattering material onto the Moon even after the detachment.

Chapter 12

The Chemistry of the Crater Interior, the Foreign Mass That Is Not There, and the Earth as the Last Scatterer →

← Chapter 10 · Prologue

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